Yashasvi Karnena

Theaterstraße 58,
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Germany
Nationality: India
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Professional Experience

Apr 23 – Present — **fka GmbH** – Aachen, Germany
Development Engineer – Software
- Developed ISO-26262-qualified real-time control software components using TargetLink and MATLAB/Simulink.
- Built Functional Mockup Units (FMUs) for validating ECU firmware against controlled plant models.
- Investigated Nonlinear control methods for PMSM position control in Steer-By-Wire (SbW) systems.
- Conducted HiL, MiL, and SiL testing of functional software.
- Researched and implemented concepts for cloud based plant parameter estimation and adaptive controller tuning of Software Defined Vehicles (SDVs).

Apr 22 – Mar 23 — **fka GmbH** – Aachen, Germany
Master Thesis Student
- Implemented position trajectory tracking controller using nonlinear 2DOF control.
- Developed nonlinear MPC trajectory planner with State-Lattice planners for solver warm-starting to tackle non-convexities.
- Proposed a state-machine based sequential motion cooperation algorithm for Connected Autonomous Vehicles.

Nov 21 – Mar 22 — **fka GmbH** – Aachen, Germany
Intern
- Designed cascaded position control loops and extended Kalman filters (Parameter Estimation) for Steer-by-Wire systems.
- Developed C/C++ estimators and filters; deployed on PXROS RTOS.
- Debugged Hardware, Firmware and control loop problems.

May 21 – Oct 21 — **Institute for Electrical Machines, RWTH Aachen** – Aachen, Germany
Student Assistant
- Created Simulink exercises for lab course – *Dynamics of Electrical Machines*.
- Implemented Simulink based bearing failure analysis models arising out of inverter harmonics.

May 18 – Apr 19 — **Altigreen Propulsion Labs Private Limited** – Bengaluru, India
Junior Electrical Engineer
- Developed firmware for telematics units and Inverter FOC algorithms in Simulink.
- Implemented toolchains using CMake/Jenkins for TI C2000 and ARM Cortex-M microcontroller platforms.

Apr 17 – Sep 17 — **Defense Avionics Research Establishment** – Bengaluru, India
Intern
- Designed power distribution boards using LTSpice and Altium.
- Implemented avionics logic on FPGAs; developed analog/digital filters.

Education

Oct 19 – May 23 **RWTH Aachen University** – Aachen, Germany
M.Sc in Electrical Engineering, IT & Computer Engineering
Major: Systems and Automation (Total Grade – 2,2)

Jul 14 – May 18 **Vellore Institute of Technology** – Vellore, India
B.Tech in Electrical and Electronics Engineering
(Total Grade – 1,9)

Skills

Programming:	C/C++, Java, Python, Golang, Bash, LaTeX, CMake
Code Generation:	MATLAB/Simulink Coder, dSPACE TargetLink, MathWorks Target Language Compiler
Simulation:	MATLAB, Simulink, LabView, PLECS, LTSpice, Simscape Language
EDA Tools:	KiCAD, Altium Designer, Eagle PCB
Testing:	CANoe, CANape, ControlDesk, PCAN-View
IT Tools:	Git, Jira, MS Office, IBM DOORS, PTC Windchill, Citavi
Languages:	English (C2), German (B1), Telugu (C2), Hindi (C1)

Professional Training

Oct 24 **Automotive Cyber Security Training** – NobleProg GmbH

Extracurricular Activities

Apr 22 – Sep 24 **Tachyon Hyperloop** – RWTH Aachen University
- Implemented Telemetry, Brakes Control and, Thermal Control hardware using Altium Designer.
- Developed high voltage power electronics package for 3-Phase propulsion control.
- Developed firmware using MATLAB/Simulink for TI C2000 microcontrollers.
- Managed finances and projects as a board member and mentored new recruits in electronics design.

Feb 15 – Sep 17 **Pravega Racing** – Vellore Institute of Technology
- Developed Electronic Throttle controller.
- Tuned Engine Control Units and built race specification wiring harness.
- Developed Traction Control Unit for race car launch control.

Awards

Aug 24 **European Hyperloop Week** – Best Electronics Package
Jan 17 **Formula Student India** – Best Engineering Design
May 13 **NASA Space Settlement Contest** – Honorable Mention

Yashasvi Karnena
Aachen, September 11, 2025